GlucoTwin

Probabilistic Physiological Digital Twin for Clinical Decision Support

Project Details & Submission

GlucoTwin is an end-to-end clinical decision support platform designed for Type 2 Diabetes (T2D) management. It continuously models continuous glucose monitoring (CGM) telemetry and electronic health record (EHR) phenotypes to forecast trajectories, detect early hyperglycemia spikes, and run counterfactual what-if simulations.

Team Details

Team Name: deepanshukuntal787

Leader: Deepanshu Kuntal

Members: Lakshya Pant, Ishan Bhardwaj

Institution: Manipal Institute of Technology

Tech Stack

Backend: FastAPI, Python, SQLite (WAL Mode)

Frontend: React, Vite, Recharts

Machine Learning: PyTorch (FusionTwinNet), Isotonic Regression

Deployment: Multi-stage Docker

Problem Statement & Healthcare Use Case

Predicting adverse clinical events requires continuous, multimodal modeling of patient physiology. However, clinicians often lack interactive tools to visualize future physiological trajectories and safely test interventions before applying them in the real world. Furthermore, the development of these machine learning systems is heavily restricted by strict data privacy laws (DPDP Act, HIPAA).

The Healthcare Use Case: GlucoTwin focuses directly on Type 2 Diabetes (T2D) management. By establishing a virtual replica of the patient's metabolic state, clinicians can preemptively detect glycemic excursions (>180 mg/dL) up to 45 minutes in advance. The conceptual UI dashboard allows doctors to simulate planned carbohydrate intake, physical activity, and stress variables to see how the patient's predicted 2-hour trajectory adapts dynamically.

Sandbox & Privacy Compliance

To navigate data constraints and strict privacy regulations, GlucoTwin was trained entirely on synthetic and anonymized datasets:

- **EHR Data:** Utilized anonymized clinical databases mimicking MIMIC-IV alongside synthetic patient generation tools mimicking Synthea to build baseline phenotypes.
- **Sensor Telemetry:** Ingested synthetic time-series data designed to emulate continuous glucose monitors and open-source wearable data (similar to Apple Health).

System Architecture

GlucoTwin operates on a highly concurrent, real-time architecture built to process 15-minute sensor ticks while serving a modern clinician-facing web interface. The system leverages WebSocket connections to stream telemetry data immediately to the dashboard.



Core Application Endpoints

- **/predict:** Generates the 2-hour trajectory forecast and alert probability.
- **/explain:** Decomposes the forecast into physiological driver attributions (e.g., meal bolus, circadian rhythm, baseline insulin resistance).
- **/whatif:** Accepts counterfactual intervention data (carbs, step count) and returns a dynamically adjusted simulated trajectory.
- **/stream (WS):** Real-time telemetry pipeline for high-frequency sensor tick replays.

ML Specifications & Project Outcomes

The core intelligence of GlucoTwin is powered by **FusionTwinNet**, a 291,140 parameter neural network. It utilizes a static EHR Multilayer Perceptron (MLP) encoder for 21 clinical features, fused with a Conv1D Temporal Stem and 2-layer Bidirectional GRU for handling dynamic 16-channel time-series data.

Performance Metric	Value	Description
AUROC	0.9978	Accuracy in predicting >180 mg/dL hyperglycemia spikes.
AUPRC	0.8544	Precision-recall metric for imbalanced excursion events.
RMSE @ 60 min	12.54 mg/dL	Root Mean Square Error of glucose trajectory at 1 hour.
MAE @ 60 min	9.81 mg/dL	Mean Absolute Error of the predicted trajectory.
ECE	0.0009	Expected Calibration Error (via Isotonic Regression).

Clinical Explainability

Forecasts are entirely interpretable. The model successfully decomposes trajectories into distinct physiological attributions, preventing "black-box" distrust from clinicians.

Intervention Monotonicity

Counterfactual modeling demonstrates a **96% compliance rate** for carbohydrate ingestion logic, and **95% compliance** for physical exertion (step count) logic.